Internet & Web. Technology

Waste Management System '24

WMS '24
WASTE NEUTRAL

WasteM'24

P R O J E C T

R E P O R T

v1.6

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FET-BDS-2022-26-020

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INDEX

Sl. No.	Section	Page No.
1	Project Overview	1
2	Database Design	1
3	User Registration & Login	2
4	Data Insertion	4
5	Reports Page	6
6	Screenshots	7
7	Future Updates	10
8	GitHub	11

1. **Project Overview**

The goal of this web application is to track and report daily management activities, ensuring users can monitor and improve their waste management practices.

- **Key Features:**

✓ **User Registration & LogIn System:** Secure account creation and login functionalities.

✓ **Daily Data Entry:** User friendly forms to input data for food waste, recycled water, solid waste, and wet waste.

✓ **Input Validation:** JavaScript validation to ensure accurate data entry.

✓ **Reporting System:** Consolidated reports displaying waste data for selected days or, a month.

2. **Database Design [** waste_management **]**

Our database consists of two main tables, capturing user information and their waste entries:

➢ users Table

- Stores user details for authentication and identification:

Column Name	Data Type	Description
id	INT (primary key)	Unique Identifier
username	VarChar	User's Chosen Alias
password	VarChar	Hashed /Encrypted Password for Security
email	VarChar	User's Email Address

➢ waste_data Table

- Records daily waste for authentication and identification:

Column Name	Data Type	Description
id	INT(primary key)	Unique Identifier
user_id	INT (foreign key)	References users table
date	date	date of entry
food_waste	int	amount of food waste (kg)
recycled_water	int	amount of recycled water (ltr)
solid_waste	float	amount of solid waste (kg)
wet_waste	float	amount of wet waste (kg)

3. User Registration & LogIn

HTML Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Register - Waste Management System</title>
    <link rel="stylesheet" href="style.css">
    <script src="script.js" defer></script>
</head>
<body>

    <header>
        <h1><a href="index.html" id="home-link">Waste Management System</a></h1>
    </header>

    <div id="content" class="container">
        <h2>Create Your Account</h2>
        <p>Please fill in the details below to register an account.</p>

        <form id="register-form" action="register.php" method="POST">
            <div class="form-group">
                <label for="username">Username:</label>
                <input type="text" id="username" name="username"
 placeholder="Choose a username" required>
            </div>

            <div class="form-group">
                <label for="email">Email:</label>
                <input type="email" id="email" name="email" placeholder="Enter
your email" required>

            </div>

            <div class="form-group">
                <label for="password">Password:</label>
                <input type="password" id="password" name="password"
 placeholder="Create a password" required>
            </div>

            <div class="form-group">
                <label for="confirm-password">Confirm Password:</label>
                <input type="password" id="confirm-password"
 name="confirm-password" placeholder="Re-enter your password" required>
            </div>

            <button type="submit">Register</button>
        </form>

        <p
 class="text-center">Already have an account? <a href="login.html">Login
here</a>.</p>
    </div>

</body>
</html>
```

JavaScript:

```
// Handle form submissions for registration
document.getElementById("register-form")?.addEventListener("submit",
function(event) {
    event.preventDefault();
    const formData = new FormData(this); // Collect form data
    fetch("register.php", {
        method: "POST",
        body: formData
    })
    .then(response => response.json())
    .then(data => {
        alert(data.message);
        if (data.success) {
            window.location.href = "login.html"; // Redirect to login page
        }
    })
    .catch(error => {
        console.error("Error registering user:", error);
        alert("Error registering user. Please try again later.");
    });
});
```

.php:

```
<?php
require 'config.php';
require 'functions.php';

ini_set('display_errors', 1);
ini_set('display_startup_errors', 1);
error_reporting(E_ALL);

header('Content-Type: application/json');

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    // Sanitize inputs
    $username = trim(htmlspecialchars($_POST['username']));
    $email = trim(filter_var($_POST['email'], FILTER_SANITIZE_EMAIL));
    $password = $_POST['password'];

    // Basic input validation
    if (empty($username) || empty($email) || empty($password)) {
        echo json_encode(['success' => false, 'message' => 'All fields are required.']);
        exit();
    }
    if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
        echo json_encode(['success' => false, 'message' => 'Invalid email format.']);
        exit();
    }
    if (strlen($password) < 6) {
        echo json_encode(['success' => false, 'message' => 'Password must be at least 6 characters
long.']);
        exit();
    }

    // Register user
    $result = registerUser($username, $email, $password);

    // Return result
    if ($result === "Registration successful!") {
        echo json_encode(['success' => true, 'message' => $result]);
    } else {
        echo json_encode(['success' => false, 'message' => $result]);
    }
} else {
    header("Location: register.html");
    exit();
}
```

Similarly, the LogIn page is programmed.

4. Data Entry Form

HTML Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Data Entry - Waste Management System</title>
  <link rel="stylesheet" href="style.css">
  <script src="script.js" defer></script>
</head>
<body>
  <!-- Header Section -->
  <header>
    <h1><a href="index.html" id="home-link">Waste Management System</a></h1>
  </header>

  <!-- Data Entry Form Section -->
  <div id="content" class="container">
    <h2>Enter Waste Data</h2>
    <p>Please fill in the details below to record your waste data.</p>
    <form id="data-entry-form" action="data_entry.php" method="POST">
      <!-- Date Field -->
      <div class="form-group">
        <label for="date">Date:</label>
        <input type="date" id="date" name="date" required>
      </div>
      <!-- Food Waste Field -->
      <div class="form-group">
        <label for="food_waste">Food Waste (kg):</label>
        <input type="number" id="food_waste" name="food_waste" placeholder="Enter
food waste quantity" required>
      </div>
      <!-- Recycled Water Field -->
      <div class="form-group">
        <label for="recycled_water">Recycled Water (liters):</label>
        <input type="number" id="recycled_water" name="recycled_water"
placeholder="Enter recycled water quantity" required>
      </div>
      <!-- Solid Waste Field -->
      <div class="form-group">
        <label for="solid_waste">Solid Waste (kg):</label>
        <input type="number" id="solid_waste" name="solid_waste"
placeholder="Enter solid waste quantity" required>
      </div>
      <!-- Wet Waste Field -->
      <div class="form-group">
        <label for="wet_waste">Wet Waste (kg):</label>
        <input type="number" id="wet_waste" name="wet_waste" placeholder="Enter
wet waste quantity" required>
      </div>
      <!-- Submit Button -->
      <button type="submit">Submit Data</button>
    </form>
  </div>
</body>
</html>
```

JavaScript:

```
// Handle form submissions for data entry
document.getElementById("data-entry-form")?.addEventListener("submit",
function(event) {
    event.preventDefault();
    const formData = new FormData(this); // Collect form data
    fetch("data_entry.php", {
        method: "POST",
        body: formData
    })
    .then(response => response.json())
    .then(data => {
        alert(data.message);
        if (data.success) {
            this.reset(); // Reset the form upon successful submission
        }
    })
    .catch(error => {
        console.error("Error submitting data:", error);
        alert("Error submitting data. Please try again later.");
    });
});
```

.php:

```
<?php
session_start(); // Start the session

require 'config.php'; // Include the database configuration
require 'functions.php'; // Include functions for data handling

// Check if the request method is POST and the user is logged in
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_SESSION['user_id'])) {
    // Retrieve data from POST request
    $date = $_POST['date'];
    $foodWaste = $_POST['food_waste'];
    $recycledWater = $_POST['recycled_water'];
    $solidWaste = $_POST['solid_waste'];
    $wetWaste = $_POST['wet_waste'];

    // Call the function to submit data entry
    $result = submitDataEntry($date, $foodWaste, $recycledWater, $solidWaste,
$wetWaste);

    // Return JSON response
    echo json_encode(['success' => $result === "Data submitted successfully!",
'message' => $result]);
} else {
    // If the user is not logged in, redirect to login page
    if (!isset($_SESSION['user_id'])) {
        header("Location: login.html");
        exit();
    } else {
        echo json_encode(['success' => false, 'message' => "You must be logged in
to access this page."]);
    }
}
?>
```

Post-DataEntry; Users can click on the URL ReDirected Window Header
' Waste Management System '
*& Navigate to the final **Reports** page.*

5. **Reports Page [** *concise & clear* **]**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Generate Waste Report - Waste Management System</title>
  <script src="script.js" defer></script>
</head>
<body>
  <header>
    <h1><a href="index.html" id="home-link">Waste Management System</a></h1>
  </header>
  <div id="content">
    <h2>Generate Waste Report</h2>
    <p>Select the type of report and fill in the details.</p>
    <form id="reports-form">
      <h3>Select Report Type</h3>
      <div class="form-group">
        <label>
          <input type="radio" name="report_type" value="date_range" checked> Date Range
          <input type="date" id="from_date" required>
          <input type="date" id="to_date" required>
        </label>
      </div>
      <div class="form-group">
        <label>
          <input type="radio" name="report_type" value="month"> Month
          <select id="month" disabled>
            </select>
        </label>
      </div>
      <button type="submit">Generate Report</button>
    </form>

    <h3>Report Results</h3>
    <table id="reports-table">
      <thead>
        <tr>
          <th>Date</th>
          <th>Food Waste (kg)</th>
          <th>Recycled Water (liters)</th>
          <th>Solid Waste (kg)</th>
          <th>Wet Waste (kg)</th>
        </tr>
      </thead>
      <tbody>
        </tbody>
      <tfoot>
        <tr>
          <td><strong>Total</strong></td>
          <td id="total_food_waste">0</td>
          <td id="total_recycled_water">0</td>
          <td id="total_solid_waste">0</td>
          <td id="total_wet_waste">0</td>
        </tr>
      </tfoot>
    </table>
  </div>
  <script>
    // Enable/disable input fields based on selected radio button
    document.querySelectorAll('input[name="report_type"]').forEach((input) => {
      input.addEventListener('change', function () {
        document.getElementById('from_date').disabled = this.value !== 'date_range';
        document.getElementById('to_date').disabled = this.value !== 'date_range';
        document.getElementById('month').disabled = this.value === 'date_range';
      });
    });  … [ MONTHs HERE! ]
    // Handle form submission
    document.getElementById('reports-form').addEventListener('submit', function (event) {
      event.preventDefault();
      const reportType = document.querySelector('input[name="report_type"]:checked').value;
      let data = { type: reportType };

      if (reportType === 'date_range') {
        data.from_date = document.getElementById('from_date').value;
        data.to_date = document.getElementById('to_date').value;
      } else {
        data.month = document.getElementById('month').value;
      }
    });
  </script>

</body>
</html>
```

6. **Screenshots**:

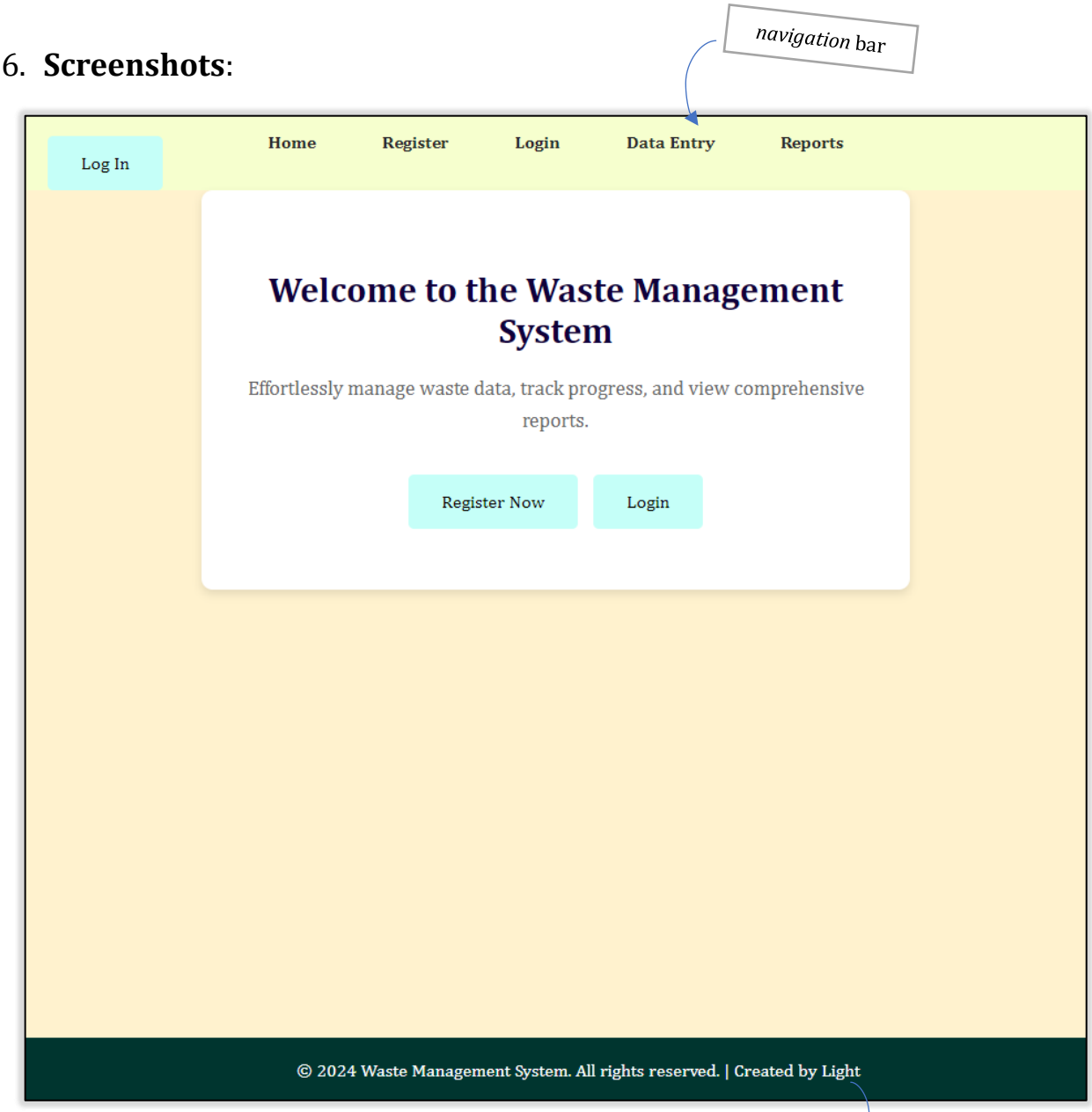


Figure 1 Home Page

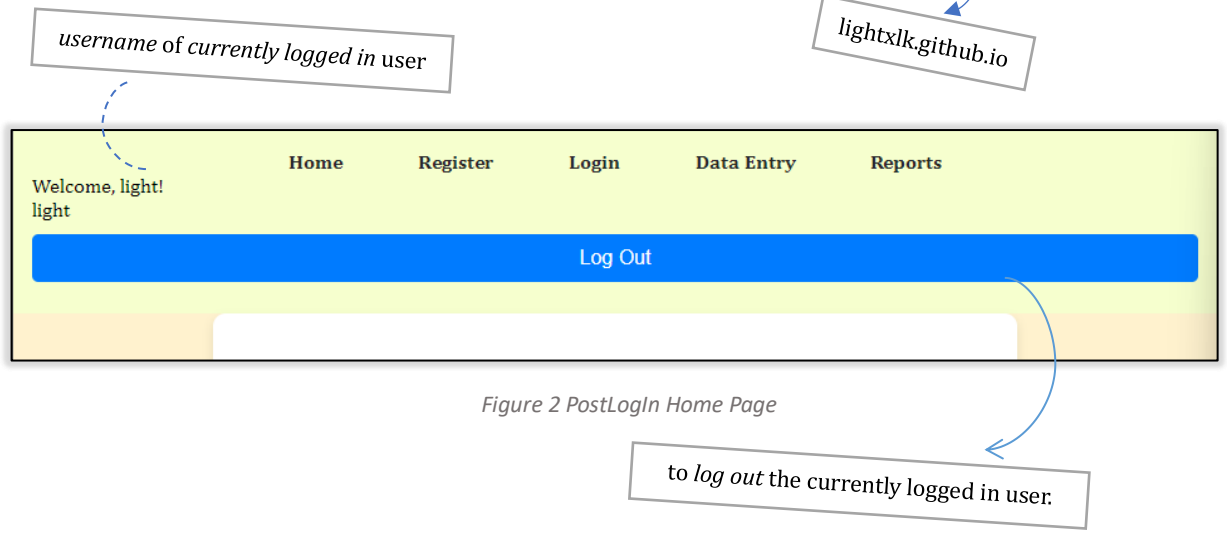


Figure 2 PostLogIn Home Page

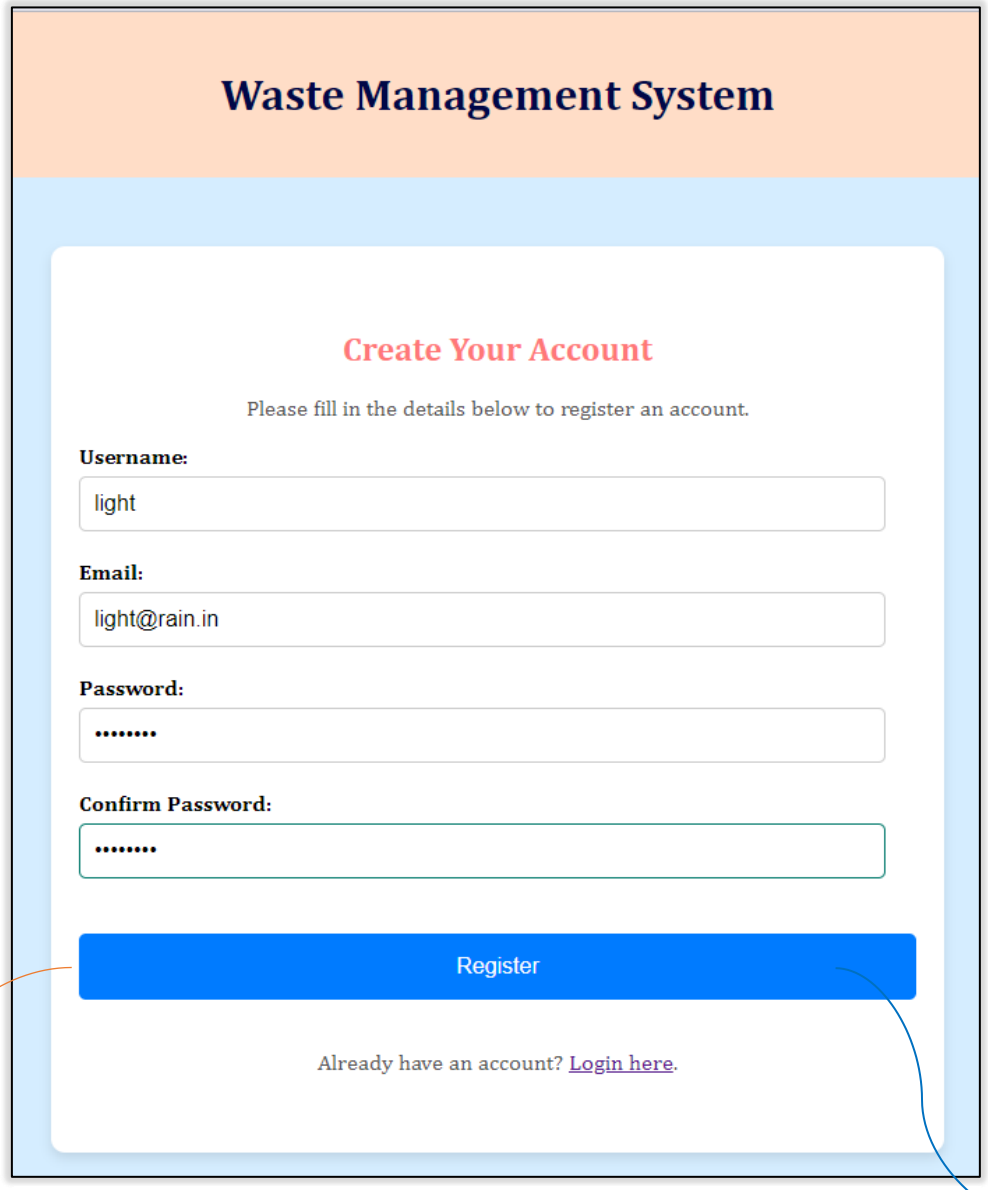
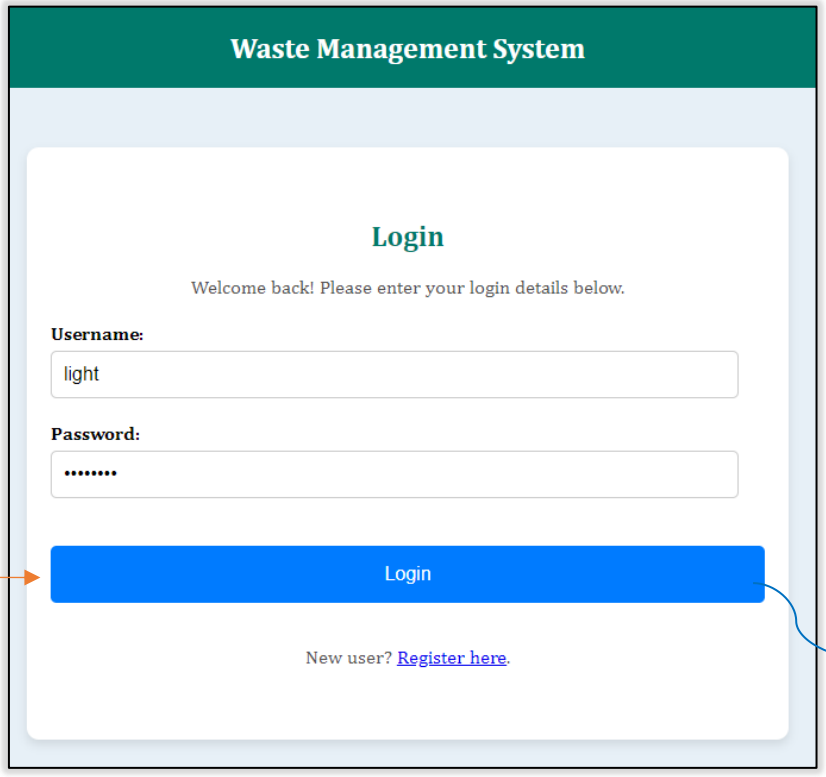


Figure 3 NewUser Registration Page

registers a *new user*, stores *user details* in the database

id	username	email	password
1	try	try@this.c.in	$2y$10$2BR5xBl1Ey7uyN
2	test	test.78@this.in	$2y$10$/q254KTCIGa7/.ak
3	sample	sample@test.in	$2y$10$NRar3ar8UJ13y7s
4	new	new@just.in	$2y$10$Uzaneg0supXiYGj
5	light	light@rain.in	$2y$10$I1K0a5vbudEPCIB



checks the *entered details* in the database & *upon validation, logs in* the user.

Figure 4 LogIn Page

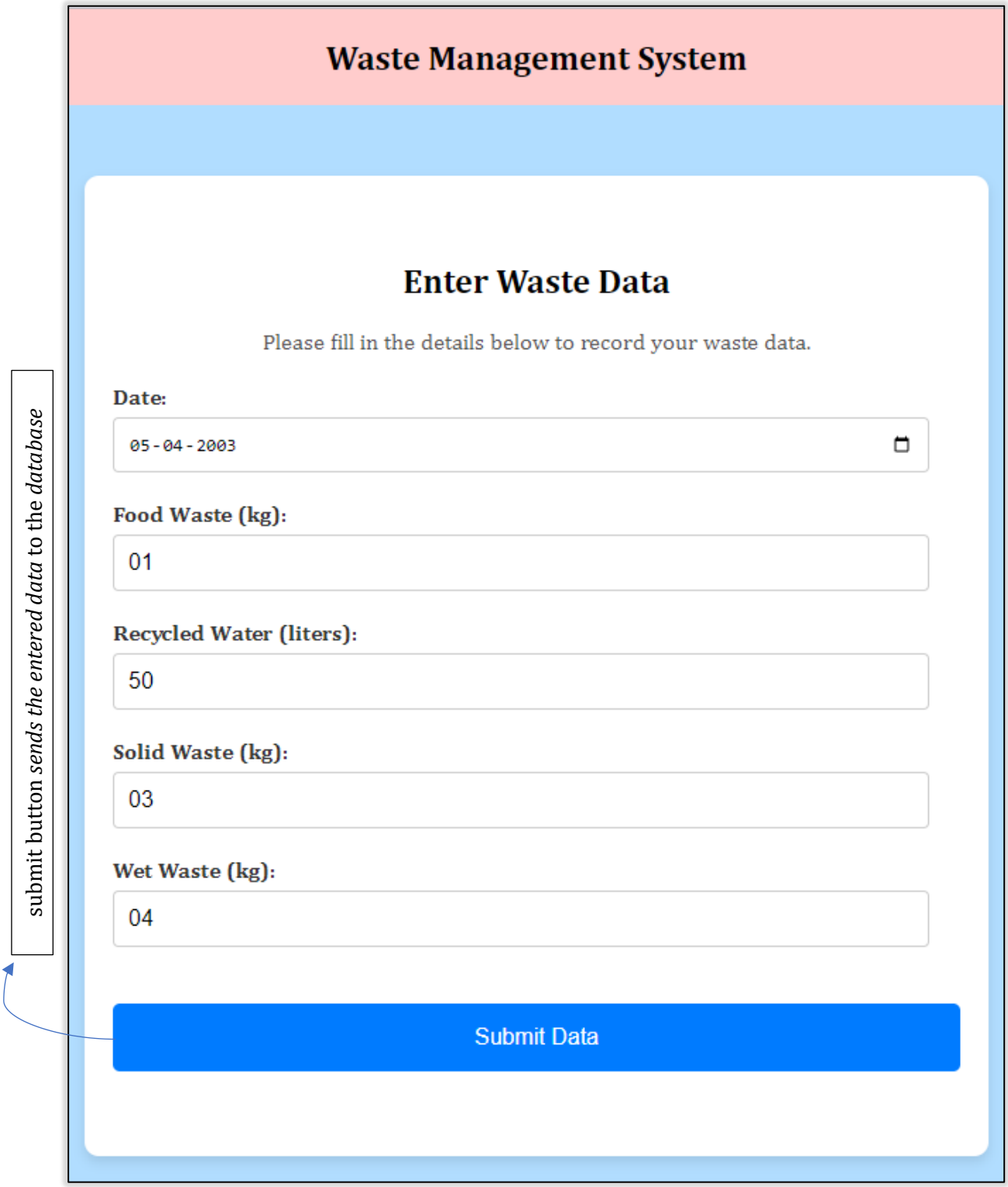


Figure 5 Data-Insertion Page

23	1	2024-10-05	89	75	46	15
24	3	2024-03-18	63	65	68	67
25	3	2024-10-12	1	2	3	1
26	4	3214-12-12	1532	651	5165	859623
27	3	2024-10-26	75	46	79	646
28	5	2003-04-05	0	10	0	0
29	5	2003-04-05	1	50	3	4

Figure 6 Inserted Data in the Database

the user entered data is inserted into this table

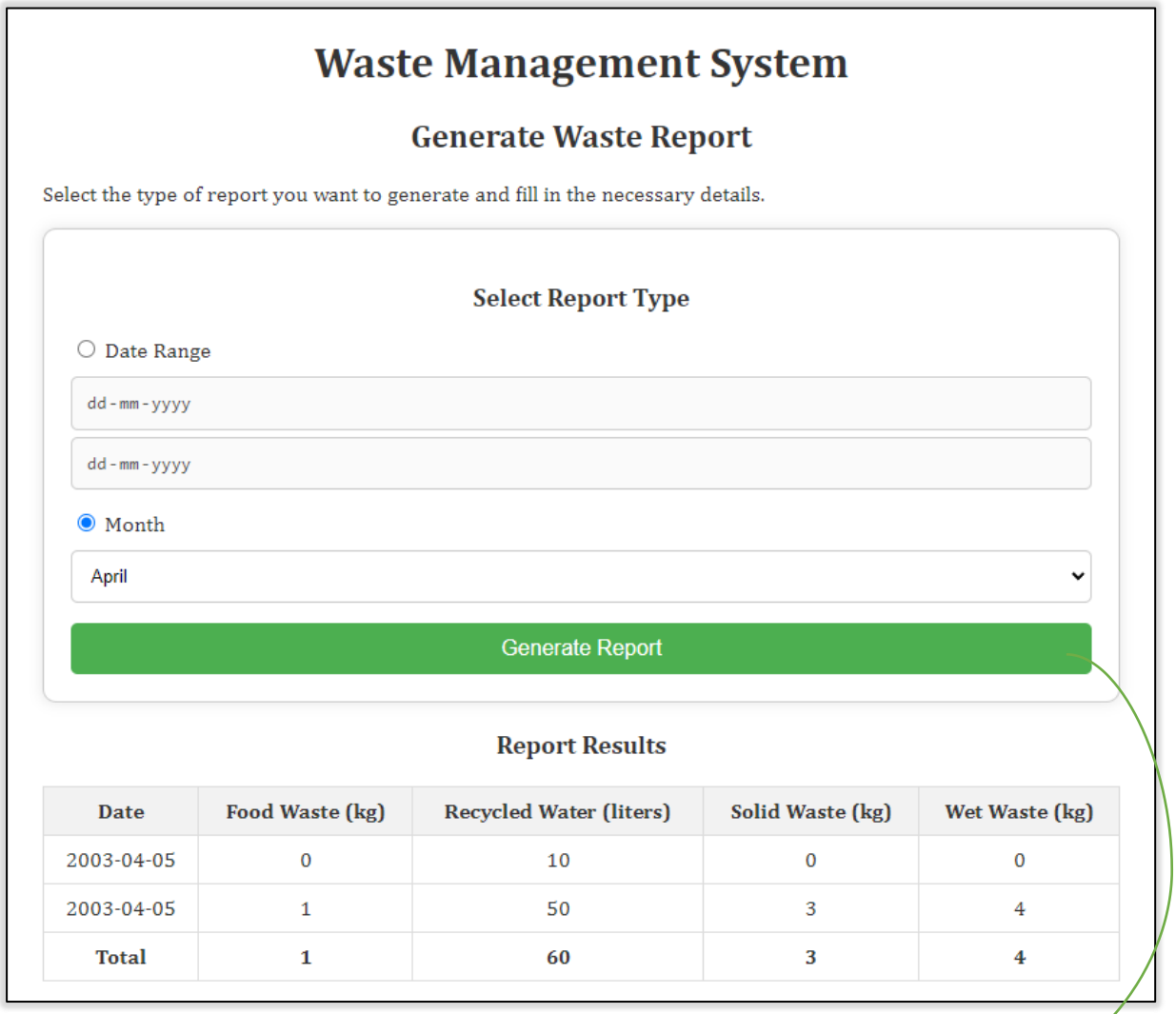


Figure 7 Reports Page

checks for the current logged in user, and then fetches their data

7. Future Upgrades:
   - Adding a *fa-eye-slash* to the **Registration** + **LogIn** Pages.
   - Adding a *dynamic picture gallery* on the **Home Page** of sustainable world.
   - Adding the **Navigation** menu to each page, so we *don't always* have to head to the **Home** Page to navigate to another page.
   - Adding **a Log Out** button to each page.
   - Adding *redirection* to **LogIn** page, in case a user tries *to access a feature without logging in*.
   - Adding a *Year feature* to the **Reports** Page.
   - Adding a *date limitation* to the **Data Insertion** and **Reports** page, to the *past and current date* only, *not* future dates.
   - Changing or, adding *more columns* to the **Database**, table for *additional waste data*, if any.
   - Update the website w/ a **favicon**.

more to be added...

8. **GitHub:**
   All files and screenshots, along with all the versions (v1 > v1.x) are uploaded and updated on GitHub under the © lightxlk [Lokesh Patra], October 2024.

Lokesh Patra

FET-BDS-2022-26-020

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